
Problem 8.21. For any positive integer x, let x^R be the integer whose binary representation is the reverse of the binary representation of x (Assume no leading 0s in the binary representation of x.) Define the function $R^+ : \mathbb{N} \longrightarrow \mathbb{N}$ where $R^+(x) = x + x^R$.

Part a. Let $A_2 = \{\langle x, y\rangle \mid R^+(x) = y\}$. Show $A_2 \in \mathrm{L}$.

Proof. □

Part b. Let $A_3 = \{\langle x, y\rangle \mid R^+(R^+(x)) = y\}$. Show $A_3 \in \mathrm{L}$.

Proof. □